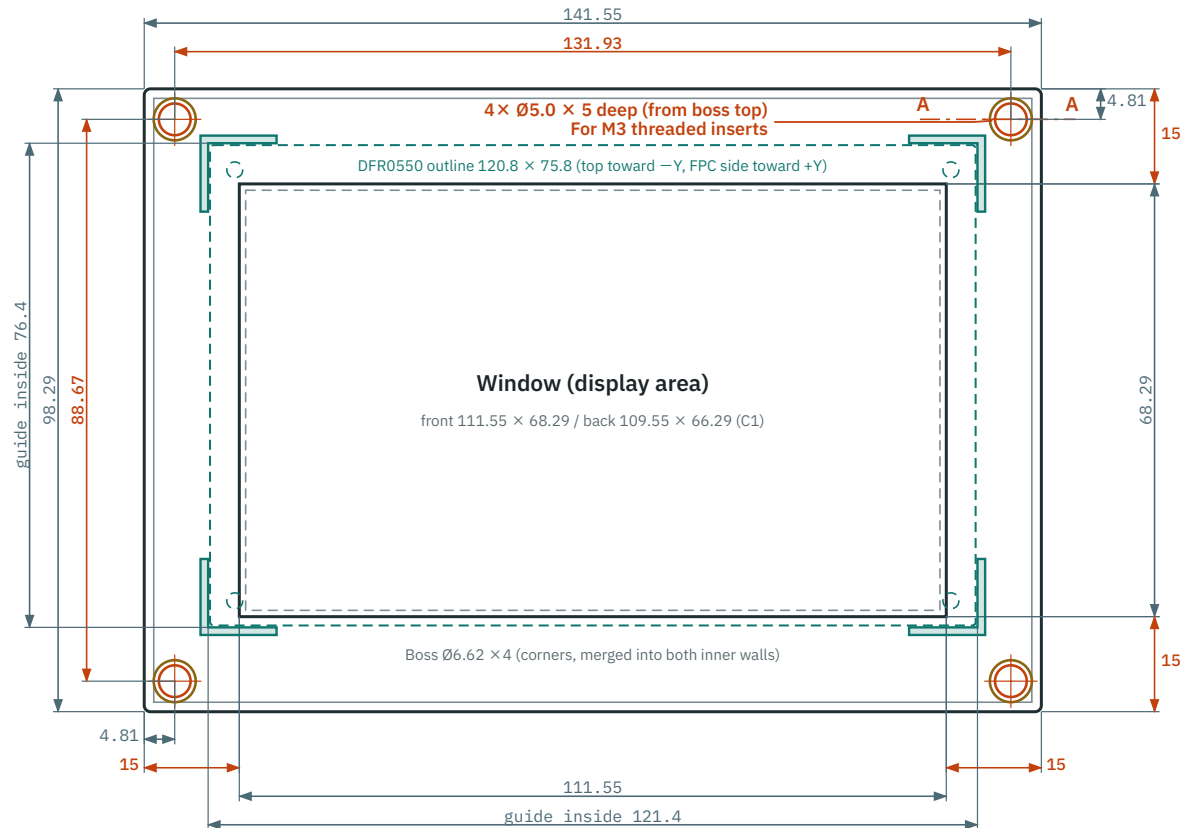


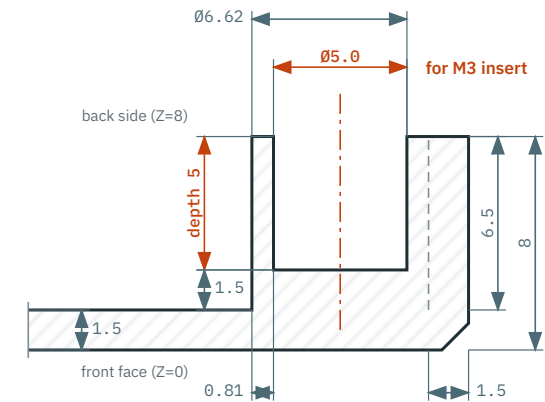
LCD Case Drawing (15 mm bezel, back cover, 1 mm panel)

Front part of the case for a 5-inch LCD. The bezel border, from the window edge to the outline as seen from the front, is 15 on all four sides. The four corner bosses take M3 threaded inserts, and the back cover screws into them. L-shaped guides locate a DFR0550. All dimensions in mm.

5_in_lcd_case_front_bezel15.scad 2026-10-07



Section A-A (top-right boss)



Ø5.0 × 5 deep holes are for M3 threaded inserts. All four bosses are the same.

The plan view is seen from the back (boss side, +Z). The dashed line in the section marks the inner wall. Coordinates match the STL / OpenSCAD model; the origin is offset from the outline center by X -1.365, Y -1.49.

Bosses and insert holes

Item	Dimension	Notes
Insert hole	4× Ø5.0 × 5 deep	For M3 threaded inserts. From boss top (Z=8) down to Z=3
Boss	Ø6.62 × 6.5 high	From plate top (Z=1.5) to Z=8. Corner side filled square and merged into the walls
Hole pitch	131.93 × 88.67	4.81 from each outer edge to the boss center
Boss centers (STL)	(67.331, 45.824) (−64.601, 45.824) (67.331, −42.844) (−64.601, −42.844)	Top right, top left, bottom right, bottom left
Material under hole	1.5	Hole bottom Z=3 to plate top Z=1.5 (3.0 to the front face)
Boss wall	0.81	On the window side. The corner side is joined to the walls

Outline

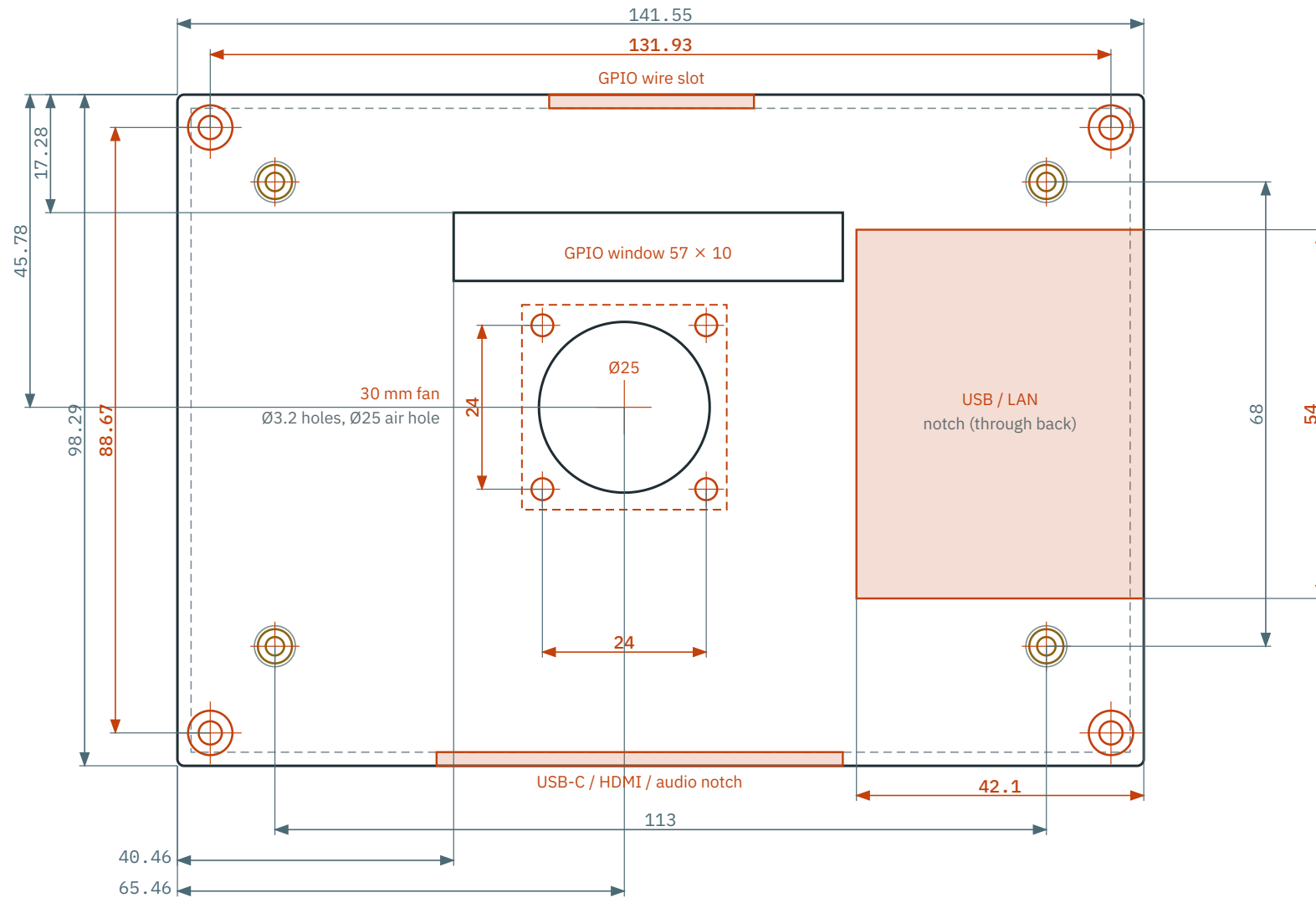
Item	Dimension	Notes
Outline	141.55 × 98.29 × 8	Corners R1, front outer edge C1
Inner wall	138.55 × 95.29	Wall 1.5 thick
Front plate	1.5 thick	
Window	back 109.55 × 66.29 front 111.55 × 68.29	C1 chamfer. From outer edges: 15 on all sides (front), 16 (back). Window center (1.365, 1.49)

DFR0550 locating guides

L-shaped guides that hold a DFRobot DFR0550 (5-inch DSI touchscreen) in line with the window. The DFR0550 mounting holes (113 × 68, M2.5) are covered by the LCD on the front, and two of them fall inside the display area, so the panel has no holes for them. Use them from the back through the standoffs.

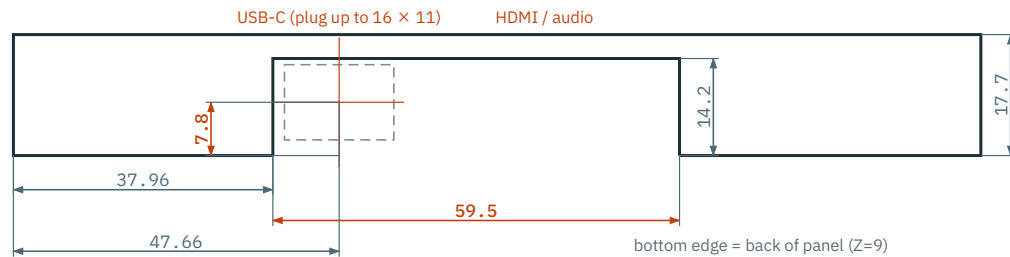
Item	Dimension	Notes
Guides	4× L, arm 12 / 1.2 thick / 4 high	At the four DFR0550 corners, from plate top (Z=1.5) to Z=5.5. Lower than the 5.7 LCD stack
Inside of guides	121.4 × 76.4	X −59.335 to 62.065, Y −34.35 to 42.05. 0.3 clearance per side around the 120.8 × 75.8 outline
DFR0550 center	(1.365, 3.85)	Center of the 113 × 68 holes. Puts the 108 × 64.8 display area in the middle of the window (about 0.75 margin)
Orientation	top → −Y	Narrow display margin (3.14) toward −Y, FPC and connector side toward +Y
FPC and connectors	27 to 94 from left	Stick out up to 1.2 from the +Y edge. Guides are only at the corners, so they clear
DFR0550 holes (reference)	X −55.135 / 57.865 Y −30.15 / 37.85	Dotted circles in the drawing. Not drilled in the panel

Back cover (mode 2) plan, seen from the back



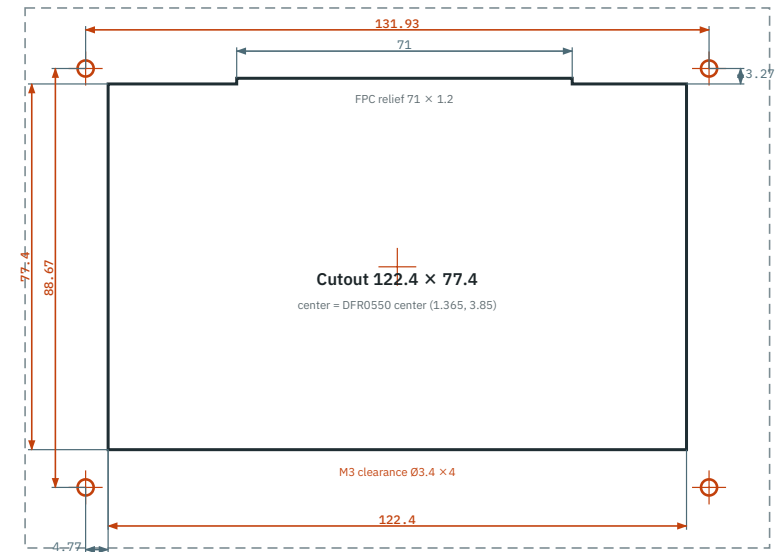
The 1 mm panel is clamped between the bezel and the back cover; M3 × 10 screws go from the back of the cover into the bezel inserts. Orange fills are wall openings. Posts at the DFR0550 outer holes press on the tips of the DFR0550 standoffs (0.2 gap, or screw in with M2.5).

—Y wall (seen from outside)



The HDMI / audio notch is extended in front of the USB-C port, so a normal USB-C cable plugs in straight from outside. The notch is open to the bottom edge (panel side), so the plug overmold (up to 16 wide, 11 high) clears it.

1 mm panel cutout (mode 4)



The back of the DFR0550 board sits 0.8 behind the panel face, so it passes through this cutout.

Back cover and assembly

Item	Dimension	Notes
Back cover outline	141.55 × 98.29 × 17.7	Same outline as the bezel. Walls 2, back plate 2. Depth set by the GPIO header
M3 screw holes	Ø3.4, counterbore Ø6.5	131.93 × 88.67. 5 under the counterbore, counterbore 12.7 deep
M3 screws (cover → bezel)	M3 × 10 (11 max)	5 under the head + 1 panel + 4 into the insert. 12 or longer hits the bottom of the 5-deep insert hole
M2.5 screws (DFR0550 hold-down, optional)	M2.5 × 6	3 post + 0.2 gap + about 2.8 into the DFR0550 standoff. Counterbore Ø5, 9.7 deep
DFR0550 hold-down posts	Ø6, hole Ø2.7, counterbore Ø5	113 × 68. Post ends 0.2 from the DFR0550 standoff tips (Z=13.8)
USB-C / HDMI / audio	59.5 × 14.2	—Y wall, open to the bottom edge. USB-C plugs in straight from outside (center X −21.755, Z 16.8, about 22 to the port)
USB / LAN	42.1 × 54 notch	Back plate and wall cut away from the +X edge (Y −23.15 to 30.85). USB/LAN stick out about 4.5 past the back
GPIO window	57 × 10	Back plate, right above the header. Plug connectors in from outside with the cover on
30 mm fan	Ø3.2 holes × 4, 24 pitch	Mounted on the outside of the back. Center (−3.955, 4.85), over the Pi4 CPU. Ø25 air hole under the fan
GPIO wire slot	30 × 6.5	+Y wall (Z 17.7 to 24.2)
Panel cutout	122.4 × 77.4	Center (1.365, 3.85). FPC relief 71 × 1.2 on the +Y edge, M3 clearance Ø3.4 × 4
Height (from front, Z=0)	Z	Notes
Bezel rim, panel front face	8	

Height (from front, Z=0)	Z	Notes
Panel back, back cover seat	9	Panel 1 thick
Back of DFR0550 board	8.8	LCD 5.7 + board 1.6 (DFRobot CAD)
DFR0550 standoff tips, Pi4 board underside	13.8	Standoffs 5
Top of GPIO header	23.7	Board 1.4 + 8.5. Tallest part inside the cover
Back face of cover	26.7	1 clearance + 2 plate
The Pi4 USB/LAN sit on the DFR0550 inner holes (58 × 43 mm) with its Board 1.4 + 1.6. Sticks out 4.5 past the back USB/GPIO notch toward +Y (check on the real hardware). SCAD mode: 1 bezel, 2 back cover, 3 assembly check, 4 panel cutout (DXF).		